[METRIC]

A-A-2758
October 18, 1991

COMMERCIAL ITEM DESCRIPTION

MARKER, TUBE TYPE

A TUBE TYPE MARKER WITH A POROUS NIB AND WATER RESISTANT INK

This Commercial Item Description is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies. This document supersedes Federal Specification GG-M-00114, which is cancelled.

This commercial item description covers tube type, porous nib markers for use in freehand marking, writing and drawing on porous and nonporous surfaces.

Types - The markers shall be supplied in one of the following types:

Type 1 - Felt, oriented fiber or porous plastic nib, chiseled tip, and ink which is water insoluble when dry.

Type 2 - Felt, oriented fiber, or porous plastic nib, conical tip, and ink which is water insoluble when dry.

1. SALIENT CHARACTERISTICS.

1.1 Color of ink - The marker shall be furnished with ink of one of the following colors: black, blue, red, green, orange, yellow, brown and purple or as specified in section 8 (Notes).

1.2 Finish - The cap or body of the marker shall be colored to identify the color of ink. The surface of the marker shall be of uniform appearance and shall be free from rough or sharp edges, burrs, blisters, scratches and other defects.

1.3 Identification Marking - The marker shall be legibly and permanently marked with the manufacturer's name or trademark and other requirements (see marking).

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

FSC 7520

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1.4 Construction - The marker shall be nonrefillable and shall operate by capillary action. The ink shall be held within the marker such that intimate contact is maintained between the nib and absorbant filler, under all conditions of foreseeable use. The absorbancy and ink retention of the filler shall insure that there shall be no free ink within the marker. The marker barrel shall be made of metal or plastic or a combination of these materials, and constructed so that the ink will emerge only through the writing nib. The marker shall be equipped with a tight fitting plastic or metal protective cap to provide protection against evaporation and leakage of the ink.

1.5 Shelf Life - The items shall have a shelf life of one year after delivery, unless specified otherwise (see section 8), and shall be capable of withstanding normal usage and handling and pass the performance tests (see section 2).

1.6 Workmanship - The marker shall be free from defects or imperfections of manufacture which may affect their durability, appearance or serviceability or present a hazard to the user.

1.7 Size - The marker, with cap in place, shall have the following dimensions:

Length: not less than 125 mm nor more than 175 mm
Diameter not less than 10 mm nor more than 20 mm.

1.8 Writing nib - The nib shall be permanently fixed within the tip to permit good control when writing and shall meet the line width requirements of paragraph 1.13.

A. The nib for a Type 1 marker shall be of felt, oriented fiber, or porous plastic material and shall be rectangular or square in cross section and shall have a chiseled tip.

B. The nib for a Type 2 marker shall be of felt, fiber, or porous plastic material and shall have a conical tip.

1.9 Startability - When tested in accordance with paragraph 2.1 the marker shall produce a continuous 200 mm line within 3 strokes after standing with the cap off for 30 minutes; within 4 strokes after standing with the cap off for 60 minutes; and within 4 strokes after standing with the cap off for 4 hours and recuperating in a horizontal position with the cap on for 24 hours. Each stroke shall be approximately 200 mm. in length.

1.10 Body and Cap Tightness - The cap shall be tight fitting but shall be removable easily without tools and shall prevent loss of ink by evaporation or leakage. When markers are tested in accordance with paragraph 2.2, the average loss of weight per marker shall not exceed 3.0 mg.

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1.11 Machine Write Test - The minimum writing distance for tube type markers, with porous nibs, when tested, in accordance with paragraph 2.3, shall be 600 meters for type 1 and 300 meters for type 2 for all colors or as specified in section 8.

1.12 Marking Characteristic - Lines drawn as described in paragraph 2.4 shall be solid and of uniform density.

1.13 Width of Marking - When the markers are tested in accordance with paragraph 2.5, the line width produced shall be:

Type 1:	Narrow Chamfer	1.5 - 3.0 mm.
	Broad Chamfer	4.0 - 7.0 mm.
Type 2:	1.5 - 2.5 mm.	

After 15 min of writing (paragraph 2.3), the line width (paragraph 2.5) shall not have increased by more than 50 percent.

1.14 Drying Time - Lines shall dry within 5 seconds on paper and within 60 seconds on aluminum when tested in accordance with paragraph 2.6. There shall be no smearing or smudging.

1.15 Light fastness - When the markers are tested in accordance with paragraph 2.7, the lines shall not show more than 10 percent decrease in reflection density for yellow, 20 percent for orange, 50 percent for purple and 30 percent for other colors. Black lines shall be exposed for 16 hours and other colors for 8 hours.

1.16 Resistance to Water - When tested in accordance paragraph 2.8, markings shall not show a marked change in color or shade and the writing paper drawn over the markings shall not cause smearing or smudging.

1.17 Impact Resistance - When the marker is tested in accordance with paragraph 2.9, there shall be no evidence of leakage. The cap and barrel of the marker shall show no damage and the writing nib shall show no signs of permanent loosening. If the nib is forced out of its normal position by the impact, it shall be restorable by straight pressure on the tip, without being distorted. The nib shall remain firm for good writing control. Any ink that is forced out around the nib shall not be sufficient to run down below the cap when the marker is held upright (nib up).

1.18 Accelerated Aging - When tested in accordance with paragraph 2.10. the marker shall not leak or show any distortion and shall pass the performance tests (see section 2).

2. TEST METHODS - Performance Characteristics Tests: Unless otherwise specified tests shall be conducted in an atmosphere of 295.15 degrees Kelvin plus or minus 2 degrees Kelvin temperature (22 degrees Centigrade plus or minus 2 degrees Centigrade) and 60 percent plus or minus 12 percent in relative humidity. All test

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paper used shall conform to UU-P-121, type III, and the basis weight shall be 32 pounds and the color shall be white.

2.1 Startability - In an ambient environment, place three unused markers in the vertical position (nib up) with the caps removed. After intervals of 30 minutes, draw the nib of each marker across a sheet of test paper in the normal writing position, three times, at a rate of approximately 100 mm/s, until a continuous 200 mm line is obtained. Normal hand pressure during this test should approximate the equivalence of 575 grams plus or minus 275 grams. The stroke lengths should traverse the test paper so as to produce 200 mm linear traces.

After an additional 30 minute cap off period in the vertical position, repeat the line drawing procedure using four strokes.

Recuperation - Using the same three markers, again position the markers nib up, without caps. After a 4 hour interval, replace the caps and allow the markers to recuperate at ambient for 24 hours in a horizontal position. At the conclusion of this period, repeat the line drawing procedure using four strokes.

2.2 Uncap and Cap - Tightness - Uncap and recap each of five unused markers to normal hand tightness 20 times, then draw a 1000 mm line with each marker on a length of machine test paper. Recap and weigh the markers to the nearest 0.1 mg and record the total weight of the five markers. Allow the markers to stand, nib up and capped for 2 hours at ambient conditions. Reweigh and record the total weight. The first weighing minus the second weighing divided by five is the average weight loss per marker.

2.3 Machine Write Test - The writing test shall be conducted on an Anja W-10B machine or equal at ambient environmental conditions with test parameters as follows:

Initial writing pressure	100 grams exclusive of the marker weight.
Writing speed	3.35 Meters per minute.
Underlay	Metal
Writing angle:	
Type 1	90 degrees with nib oriented to produced widest line.
Type 2	65 degrees plus or minus 5 degrees.
Test paper	Performance Testing Paper: UU-P-121, type III and the basis weight shall be 32 pounds and the color shall be white.

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Prior to the test, a small mass tone of abutting, but non-overlapping strokes, shall be drawn manually for initial reflection density readings, using each marker.

Conduct machine write tests in 60 meter increments. Immediately upon reaching each 60 meter interval, contact between nib and paper shall be interrupted and the markers capped. Markers shall be allowed a 15 minute horizontal recuperation and mass tones drawn thereafter for reflection density readings at each interval. Writing pressure shall be increased by 10 grams at each interval.

Continue 60 meter interval write test sequencing until the mass tone is of light intensity as determined by a 30 percent decrease in reflection density or until the machine pattern becomes irregular or incomplete, which ever comes first. The last acceptable 60 meter interval establishes the cumulative write-out distance.

2.4 Marking Characteristic - Draw lines on masonite (smooth side), test paper, smooth glass, and smooth tin or aluminum panels, and examine the lines for uniformity of intensity.

2.5 Width of Marking - Draw lines on glass or transparency film with the markers held in a vertical position under a weight of 575 grams plus or minus 275 gram range. Measure the width of the lines microscopically. The lines shall be drawn for type 1 in both the narrow and broad orientation.

2.6 Drying Time - Draw five lines at least 150 mm long on test surfaces of test paper and 24ST Alcald aluminum. The lines shall be drawn, for type 1, in both the narrow and broad orientation. Allow to dry for 5 min. on paper and 1 min. on aluminum. Immediately after the specified drying time, place a sheet of test paper over the lines and place a 200 gram mass approximately 30 mm in diameter on the sheet near one edge. Draw the sheet with the mass perpendicularly across the center of the lines at a rate of approximately 100 mm/s. Examine the test surface for smearing and smudging. A slight trace of color transferred to the top sheet and a slight striation of the line on the aluminum panel are permissible.

2.7 Light Fastness - Draw a series of lines as close together as possible but not overlapping on a 60 x 150 mm panel of test paper, parallel to the long dimension, and allow to dry for 5 minutes. Determine the reflection density of the lines with a densitometer, using a Wratten 106 filter or equivalent, and the instrument zeroed with a magnesium oxide standard. Use at least 10 sheets of test paper as backing for the panel during the reflection density measurements. Take at least 10 readings and average the results. Expose the panel for the specified time (actual machine exposure) in accordance with ASTM G-23 (Method 3 - continuous exposure without water spray). After exposure, determine the reflection density of the lines in the same manner

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as before. Calculate the percentage decrease in reflection density as:

$$\frac{\text{(Density before exposure - Density after exposure) x 100}}{\text{Density before exposure}}$$

2.8 Resistance to Water - Draw lines on test paper, and allow to dry for 24 hour. Immerse the paper with the line markings in water at room temperature to a depth corresponding to two-thirds of the height of the markings for 4 hours. Remove the paper with the markings from the water, allow it to dry completely, and examine the markings for any change in color or shade, Test the markings for smudging in accordance with paragraph 2.6.

2.9 Impact Resistance - Drop three capped markers 3 times each on a wooden floor from a height of 1.5 meters, first on the end opposite from the nib, second on the side, and third on the cap. Examine the body and cap for cracks and leakage, and the nib for loosening and leakage. Draw a few sharp turns, reversals and circles, and examine the nib for signs of loosening.

2.10 Accelerated Aging - With three unused markers draw a line approximately 200 mm long with each marker, on a sheet of test paper. Cap the markers tightly and place them in an air circulating oven at 60 degrees C plus or minus 1 degree C for 7 days in the horizontal position. Remove the markers from the oven and examine for leakage or distortion. Place the markers on their side and allow to cool at room temperature for 24 h.

3. REGULATORY REQUIREMENTS.

The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

4. HAZARDOUS MATERIAL REQUIREMENTS.

4.1 Toxicity - Use of any toxic substance must be in accordance with all applicable federal statutes. The use of any known carcinogens as ingredients is prohibited. The harmful substances shall be defined as the poison and injurious substances as designated in Fed Std. 313. The vendor shall prepare a certification that the inks conform to the above requirements.

4.2 Material safety data sheets - Safety data shall be furnished in accordance with Fed. Std. 313. The pertinent government mailing addresses for submission of data are listed in appendix B of Fed. Std. 313.

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4.3 Labeling - Containers shall be labeled to comply with the "Federal Hazardous Substances Act Regulations", 16 Code of Federal Regulations, Part 1500.

5. QUALITY ASSURANCE REQUIREMENTS.

5.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order (see section 8), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

5.2 Lot definition - Each lot of markers shall be inspected for conformance to the requirements herein. Inspection shall be in accordance with MIL-STD-105 using the sampling plans specified below. The lot shall be expressed in markers manufactured from the same materials, at essentially the same time.

5.3 Examination of the end item for visual defects - The sample unit for this examination shall be one marker. A random sample shall be taken from each lot in accordance with Inspection Level S-3. The AQL shall be 2.5 percent defective. Samples shall be examined for defects listed below. Any marker containing one or more defects shall be rejected.

Examine	Defects
Workmanship	See Workmanship paragraph
Material	Not as specified

5.4 Examination of the end item for dimensional defects - The sample unit for this examination shall be one marker. A random sample shall be taken from each lot in accordance with Inspection Level S-2. The AQL shall be 4.0 percent defective. Samples shall be examined for defects listed below. Any marker containing one or more defects shall be rejected.

Examine	Defects
Diameter	Not as specified
Length	Not as specified

5.5 Sampling of end items for performance tests - Sampling for performance testing shall be performed in accordance with MIL-STD-105. A random sample from each lot in accordance with Inspection Level shall be S-2 with an AQL of 4.0 percent defective. The sample unit shall be one marker.

5.6 Packaging, Packing, and Marking - Inspection Level S-2 with an AQL of 4.0 percent defective. The sample unit shall be

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one fully prepared shipping container inspected for conformance with the requirements of section 6.

6. PREPARATION FOR DELIVERY

6.1 Unless otherwise specified (see section 8), preparation for delivery shall conform to normal commercial practice.

6.2 All packages shall be marked to show month and year of manufacture (date of marker assembly).

A. Packaging, and packing - Packaging, and packing shall be as indicated below, unless specified otherwise in the contract or purchase order (see section 8). The normal commercial number of unit packages shall be packed in a close-fitting fiberboard box used commercially for this item. The complete pack shall be in compliance with the Uniform Freight Classification and the National Motor Freight Classification. The weight of the shipping container shall not exceed 65 pounds.

B. Marking - Marking shall be as indicated below, unless specified otherwise in the contract or purchase order (see section 8).

1. For Shipment To Civil agencies - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123. Bar code marking is required, unless specified otherwise (See section 8).

2. For Shipment To Military activities - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Bar code marking is required, unless specified otherwise (See section 8).

3. Additional marking - The unit and shipping containers shall include the date of manufacture and the date of reinspection (12 months after the date of manufacture) and the shipping containers, in addition, shall include any precautionary marking required by the National Motor Freight Classification Item 222, Uniform Freight Classification Rule 41, and Department of Transportation regulations.

7. APPLICABLE DOCUMENTS.

The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

7.1 Public Law and Code of Federal Regulations:

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Code of Federal Regulations and Federal Acquisition Regulations are for sale by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

Public Law 94-580 - Use of Recovered Materials

16 CFR Part 1500 - Federal Hazardous Substances Labeling Act.

7.2 National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W. Washington, D.C., 20036.

7.3 Uniform Classification Committee Agent:

Uniform Freight Classification.

Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.

7.4 Federal Standards, Specifications, and Commercial Item Descriptions:

Federal standards and specifications are available from GSA Business Service Centers and Specification and Consumer Information Distribution Section, Building 197, Washington Navy Yard, Washington DC 20407.

Single copies of this specification and other specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Washington.

FED-STD-123 Marking for Domestic Shipment (Civil Agencies)

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

UU-P-121 Paper, Bond and Writing: White and Colored.

7.5 Military Standards:

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Military Standards are available from the Standardization Documents Order Dept. Bldg. 4D 700 Robbins Avenue, Philadelphia, PA 19111-5094.

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

7.6 American Society for Testing Materials (ASTM):

G-23 Standard Practice of Operating Light Exposure Apparatus (Carbon Arc Type) With and Without Water for Exposure of Nonmetallic Materials.

ASTM standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

8. NOTES.

Ordering information - Purchasers should select the required options permitted herein, and include the following information in procurement documents:

A. Title, number, and date of this commercial item description.
B. Purchasers shall indicate type of tube marker and color of ink.
C. Purchasers shall specify arrangement for inspection and inspection facilities, if other than specified
D. Purchasers shall specify packaging, packing, and any special marking for shipment, if other than required.
E. Purchasers shall specify identification marking, if other than required.
F. Purchasers shall specify if bar code marking is not required.
G. Purchasers shall specify MSDS's required, in addition to those required by Fed-Std-313.
H. Purchasers shall specify palletization, if required.
I. Purchasers shall specify a shelf life, if other than required.
J. Purchasers shall specify write-out distance, if other than required.

MILITARY INTERESTS:

NONE: DOD has determined that no military activity has an official interest in this commercial item description.

Preparing Activity:
GSA-FSS

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